

OSSAMA TAHA

BI & DATA ANALYTICS ENGINEER

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SUMMARY

BI & Data Analytics professional with a B.Com in Accounting and 2+ years of experience building production-grade data pipelines, automated reporting systems, and executive dashboards. Proven track record of reducing reporting cycles by 10x through end-to-end automation, enabling data-driven strategic decisions for sales and operations teams. Combines deep business domain knowledge with strong technical skills in Python, SQL, PostgreSQL, Apache Airflow, dbt, and cloud infrastructure.

EXPERIENCE

Nile for Cables and Electronics, BI & Sales Data Analyst

Giza, Egypt

- Reduced reporting cycle from days to hours by designing and deploying a semi-realtime automated reporting system, directly improving sales team responsiveness and strategic decision-making Nov 2024 – present
1 year 7 months
- Led quarterly sales performance evaluations using consolidated KPIs, providing actionable insights that drove revenue-focused business decisions
- Built executive dashboards supporting sales forecasting and real-time KPI monitoring, enabling stakeholders to track performance without manual data pulls
- Designed end-to-end ETL pipelines using PostgreSQL and Dolt (version-controlled SQL database), ensuring data integrity and audit trail for critical financial metrics
- Developed a local AI chatbot using open-source LLMs for natural language querying of business data, empowering non-technical stakeholders to self-serve analytics

Samsung Innovation Campus, Data Engineer

Giza, Egypt

- Developed scalable data processing workflows using Apache Spark and Kafka, supporting business intelligence and analytical use cases July 2024 – Oct 2024
4 months
- Performed exploratory data analysis and feature engineering to enable data-driven business insights
- Deployed production data workflows on AWS cloud infrastructure (EC2, S3), ensuring reliable and cost-efficient operations

PROJECTS

dbt Data Warehouse — Star Schema & Analytics Layer

- **Problem:** Raw operational data lacked dimensional structure for efficient analytics and BI tool consumption, forcing manual data wrangling for every report
- **Process:** Designed star schema with fact and dimension tables, built 15+ dbt models across staging, intermediate, and marts layers with automated testing and documentation
- **Solution:** Production-grade data warehouse with SCD Type 2 tracking, incremental processing, and 99.9% data quality via dbt tests — enabled self-service analytics for business users
- **Business Impact:** Reduced query complexity by 70%, eliminated manual data preparation, enabled analysts to answer business questions directly

Atlas Analytics Platform

- **Problem:** Enterprise needed a unified analytics platform for multi-department data visibility, but existing solutions were siloed and required manual updates
- **Process:** Built full-stack platform with Python backend, automated ETL pipelines, and role-based interactive dashboards; deployed on AWS with CI/CD
- **Solution:** Production system with automated data sync, real-time reporting, and controlled access across departments
- **Business Impact:** Enabled data-driven decisions across 3 departments, reduced manual reporting effort by 80%, improved data accuracy and timeliness

Airflow ETL Pipeline — Multi-Source Data Orchestration

- **Problem:** Manual data extraction from multiple APIs was error-prone and lacked scheduling, monitoring, or failure recovery
- **Process:** Designed 4 production DAGs with task dependencies, dynamic task mapping, XCom data passing, and automated retry logic using Apache Airflow and Docker
- **Solution:** End-to-end orchestrated pipeline extracting from 3 API sources, transforming in staging, and loading into PostgreSQL with real-time monitoring dashboard
- **Business Impact:** Fully automated daily and weekly data ingestion with zero manual intervention, complete visibility into pipeline health

Agentic AI Workflow — Multi-Agent DataOps Automation

- **Problem:** Data operations required constant manual monitoring, debugging, and intervention across multiple systems
- **Process:** Built multi-agent system with orchestrator pattern, 5 specialized agents, tool integration, and persistent memory using LangChain and LangGraph
- **Solution:** Natural language interface for DataOps — agents autonomously monitor pipelines, run quality checks, and debug failures without human intervention
- **Business Impact:** Reduced mean-time-to-resolution for pipeline failures by 70%, automated 90% of routine monitoring and maintenance tasks

AWS Cloud Data Pipeline — Serverless ETL with IaC

- **Problem:** On-premise data infrastructure lacked scalability, reliability, and cost efficiency for growing data volumes
- **Process:** Designed S3 data lake architecture with Glue ETL, Athena serverless queries, and Step Functions orchestration using Terraform for infrastructure-as-code
- **Solution:** Fully serverless pipeline deployable with a single command — auto-scaling, zero server management
- **Business Impact:** 90% cost reduction vs traditional infrastructure, sub-second query performance on analytical data, full infrastructure version control

Kafka Streaming Pipeline — Real-Time Data Processing

- **Problem:** Batch processing couldn't meet real-time analytics requirements for stock monitoring and operational telemetry
- **Process:** Built event-driven architecture with Kafka producers, consumer groups, windowed aggregation, and dead letter queues for fault tolerance
- **Solution:** Real-time pipeline processing 1000+ events per second with live dashboard, automated alerting, and PostgreSQL persistence
- **Business Impact:** Sub-second data latency, automated threshold alerting, real-time operational visibility across monitored systems

SKILLS

- BI & Data Visualization:** Power BI, DAX, Tableau, Excel, Streamlit, Plotly, Dash
- Data Pipeline & Architecture:** ETL/ELT Pipelines, Apache Airflow, dbt, Apache Spark, Kafka, Data Modeling (Star Schema), PostgreSQL, MySQL
- Cloud & Infrastructure:** AWS (S3, Glue, Athena, Lambda, Step Functions, EC2), Terraform, Docker, Docker Compose, Linux Server Administration, Nginx, GitHub Actions (CI/CD)
- Programming & Scripting:** Python, SQL, Bash, JavaScript, TypeScript, HTML/CSS
- AI & Machine Learning:** LangChain, LangGraph, Hugging Face Transformers, Scikit-learn, TensorFlow, NLP, Local LLM Deployment
- Data Quality & Monitoring:** Great Expectations, Data Validation, Schema Testing, Anomaly Detection, Pipeline Monitoring
- Automation & Tools:** Git, n8n, OpenClaw, KiloCode, Hermes, Dolt, Jupyter, VS Code

EDUCATION

- Al-Azhar University**, Accounting & Business Management Sept 2019 – June 2023
Bachelor of Commerce
- Self-taught in programming, data engineering, and cloud technologies through professional certifications and hands-on projects

CERTIFICATIONS

- Google Data Analytics Professional Certificate**
- Data cleaning, analysis, visualization with spreadsheets, SQL, Tableau, and R
- IBM Data Engineer Professional Certificate**
- ETL pipelines, data warehousing, big data (Spark, Hadoop, Kafka), cloud deployment
- IBM Data Science Professional Certificate**
- Machine learning, data analysis, Python, SQL, applied data science capstone projects
- Samsung Innovation Campus Big Data Engineering Certificate**
- "Top Achiever (#1) in the Big Data Engineering track"
 - Intensive training in big data infrastructure, distributed computing, and data engineering
- AWS Certified Data Engineer – Associate**
- S3, Glue, Athena, Lambda, Step Functions, and Terraform (In Progress)
- iMBA Business Analysis Specialization (In Progress)**
- Coursera iMBA from UIUC – business process modeling, requirements engineering, data-driven decisions

LANGUAGES

Languages: Arabic: Native | English: Professional | Russian: Basic